A Survey of Learning-Based Methods for MRI Brain Tumor Analysis

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***Abstract*—The magnetic resonance imaging (MRI) of brain tumors in the evaluation of brain tumors is of paramount importance in diagnosis and the planning of treatment of the tumor. The recent developments in the learning-based approaches have facilitated more accurate and robust detection, classification, and segmentation of tumors. Nevertheless, numerous current methods are based on a single task or a particular architecture, which restricts their integration in clinical settings and real-time functionality. This questionnaire provides an organized overview of machine learning, deep learning, transfer learning, CNN-Transformer hybrid model, and explainable AI models of MRI-based brain tumor analysis. The reviewed articles are carefully classified according to the learning paradigm, architectural design, purpose of task, and clinical importance. The comparison analysis reveals the methodological tendencies, development of the performance, and practical constraints in widespread datasets and evaluation measures. The major issues, such as the scarcity of annotated data, model interpretability, reliability, and deployment issues, are highly discussed. Lastly, the future directions of the research are described with the focused research on unified multi-task frameworks, uncertainty-sensitive segmentation, lightweight real-time systems, and clinically trustworthy explainable models.**

Keywords— Brain tumor detection, MRI, Brain tumor detection, Brain tumor Segmentation, brain tumor detection, Transformer learning, CNN, ML

I. INTRODUCTION

Over Brain tumor is one of the most severe neurological conditions, which is complicated in morphology, with a heterogeneous tissue structure and has an important influence on the functioning of the brain. Proper identification and demarcation of tumor areas are required in diagnosis, treatment planning as well as in postoperative monitoring. The imaging modality that is most suitable in the analysis of brain tumor is Magnetic Resonance Imaging (MRI), which has the best soft-tissue contrast, multi-modal and non-invasive imaging. Nevertheless, manual analysis of MRI images is not only time-consuming but also subjective and highly reliant on radiological experience, and it is quite challenging to conduct a consistent large-scale clinical evaluation. In order to mitigate these drawbacks, learning based approaches have been widely studied with regards to automated tumor detection, classification, and segmentation. The initial machine learning (ML) models used hand-coded features extraction methods along with conventional classifiers like the Support Vector Machines, Random Forests, and the k-Nearest Neighbors. In spite of promising results of these techniques, their functionality was highly reliant on the engineering of features and it was frequently not robust against tumors that had a considerable degree of variability in size, shape and intensity. With the introduction of deep learning (DL), the analysis of brain tumors using MRI has changed dramatically because it is now possible to learn the features end-to-end using raw imaging data. Convolutional Neural Networks (CNNs), U-Net variants and attention-based architectures have also shown significant progress in classification as well as segmentation tasks. More recently, transfer learning and CNN-Transformer hybrid models have been proposed in order to learn contextual dependencies across the globe and improve representation abilities. Regardless of such improvements, there are still a number of issues, such as unavailable annotated datasets, overconfidence of the model, interpretability limitations, computational complexity, and the possibility of deploying it in a clinical setting in real time. Explainable artificial intelligence (XAI) and uncertainty-aware learning strategies have become a more and more popular topic to enhance clinical trust and reliability. The goals of saliency-based visualization, calibration, and probabilistic modeling are to improve the level of transparency and resiliency in safety-critical medical computer programs. This change is indicative of a larger change of accuracy-based research to clinically reliable and deployable intelligent systems. Though many surveys have considered either classification or segmentation techniques alone little has been done to adopt integrated systems that incorporate segmentation, classification, prognosis modeling, explainability, and real-time deployment concerns in an organized taxonomy. In addition, comparative studies involving the learning paradigms and architectural trends are still limited. Through this survey, these gaps are addressed through a systematic and detailed review of learning-based techniques of MRI brain tumor analysis. The key findings of this work can be concluded as follows:

An organized taxonomy of classical ML, deep learning, transfer learning, hybrid CNN-Transformer models, explainable AI models and multi-task models.

Critical comparative analysis with emphasis on methodological development, performance patterns, strong and practical weaknesses among datasets and measures of evaluation.

Recent challenges and future research opportunities that focus on the unified multi-task systems, uncertainty-aware segmentation, lightweight real-time architectures, and clinically trustworthy explainable models have been identified.

II. LITERATURE SURVEY

According to the recent study of MRI-based brain tumor image analysis, the traditional machine learning (ML)-based models have become complex deep learning (DL), combined models, and explainable AI (XAI) models. The initial research mainly relied on handcrafted texture, shape and intensity features with classical classifiers. As an instance, [15] proposed a hybrid system with various texture feature extractors, including Gabor 2D, Haralick, Local Binary Pattern (LBP) and spectral features with the application of ML classifier and the highest accuracy of 99.4% on Figshare MRI dataset was achieved. Equally, [1] and [6] used feature-engineered ML pipelines using optimized classifiers with a top accuracy on public and Kaggle MRI datasets of up to 99%. Though possible, they were sensitive to feature selection, and could be weak to tumor heterogeneity.Along with the advent of deep learning, optimized CNN architectures increased automated feature extraction and classification by a significant margin.Hypbrid segmentation-classification networks and ensemble CNN models in [4], [7] were more rob-tough in terms of diagnoses, and an optimized CNN in [14] was 97.03% accurate. Transfer learning was useful in small datasets; fine-tuning VGG and ResNet models in [8] achieved about 98% and the EfficientNet-B4 model in [12] achieved 99.66% accuracy on the Br35H dataset. More advances in transfer learning were made towards better feature representation, such as [24].XAI methods are added to make the DL models more interpretable. Grad-CAM on ResNet50 in [5] and [9] gave 98.52% accuracy on visual explanations, whereas ensemble and SHAP-based models in [23] and [25] improved model transparencyThe latest developments include CNN-Transformers hybrids, attention models, and multimodal combinations, which have been discussed in [3], [10], [11] and in detail in [18].Segmentation models that used transformer-based and 3D-attention-based models performed well at the benchmark level (Dice = 0.912 in [19] and Dice = 0.916 in [20]). The frameworks that were able to predict prognosis in multi-tasking were suggested in [17] and [22], and lightweight and real-time systems were constructed in [16] and [2]. In [21], the joint learning framework between mobile attention and XAI recorded an equivalent performance in concurrent segmentation and classification.All in all, the literature shows that there is a gradual transition to explainable, Transformer-enhanced, and multi-task DL systems. Though the improvement of performance is significant, majority of the studies have been task-oriented, and additional studies are necessary to combine segmentation, classification, prognosis modeling, explainability, and real-time deployment into one clinically viable framework. framework

III . METHODOLOGICAL TAXONOMY

To offer a systematic comprehension of the available methods, the studies reviewed are divided into categories based on the learning paradigm, the architectural design, the task objective, and the applicability to the clinic. Fig. 3.1 depicts the general workflow of the MRI-based analysis of brain tumors where the multi-modes of MRI are acquired (T1, T2, FLAIR) then processed by pre-processing (skull stripping, intensity normalization). The learned images are then subjected to learning-based models, such as classical machine learning, deep learning, transfer learning, hybrid attention-based architecture, and multi-task architecture.

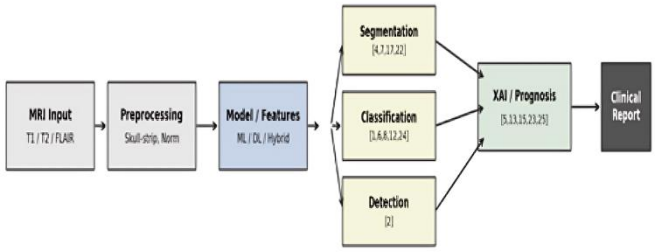

Fig 3.1 Overall Workflow of Brain tumor Analysis

The learned representations are used in three major branches of tasks segmentation, classification, and detection. Segmentation-oriented models [4], [7], [17], [22] are developed to define subregions of the tumor, and classification models [1], [6], [8], [12], [24] are used to predict the type of tumor. Detection based frameworks [2] focus on real-time localization. Sophisticated systems also combine explainable AI (XAI) and prognosis modeling [5], [9], [15], [23], [25] to improve the clinical interpretability and decision support. The last step is to produce clinically meaningful reports, which intervenes between algorithmic predictions and actual diagnostic processes. This taxonomy shows how individual task specific models have evolved into integrated, understandable, and deployable systems.the dataset is not one-sided and can be used as a predictor.

A. Classical Methods and Machine Learning:

Classical machine learning (ML) methods to process MRI-based brain tumor are based on handmade feature extractions and traditional classifiers. Characteristic features that are usually extracted are texture specialties (e.g., GLCM, LBP), intensity-based features, shape features, and statistical features obtained on the basis of tumor areas. These attributes are then passed into the classifier which can be Support Vector Machines (SVM), Random Forests (RF), k-Nearest Neighbors (kNN). They are computationally efficient and especially appropriate in situations where training data is limited or where there are limited computational processes. According to representative studies [1], [6], with the help of good feature engineering and classifier optimization, a classification accuracy of around 97% and 99% was reported respectively.

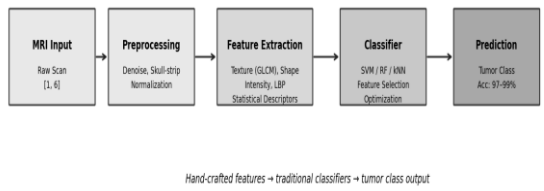

Fig 3.2 Classical ML methods approach

Fig 3.2 depicts the overall process diagram of classical ML methods, i.e. preprocessing, feature extraction, classifier training and prediction.

B. Deep Learning Approach:

The power of deep learning (DL) models, which can learn hierarchical and discriminative representations based on the raw imaging data, has enabled them to become the new paradigm of MRI-based brain tumor analysis. CNNs, U-Net based networks, and encoder-decoder networks are commonly used in classification and segmentation. Those frameworks based on segmentation [4], [7] combine tumor delineation with downstream classification to improve the diagnostic performance with reporting classification accuracies ranging between 96-97%. Architectures based on U-Nets [17], [22] also provide the capability of multi-task learning and allow segmentation, classification, and prognosis estimation to be performed in a single design. Further, lighter attention based CNN models suitable to resource limited or edge conditions [16] are found to perform competitively (around 98% accuracy) with less computational cost and hence are suitable in real-time clinical use.

Fig 3.2 Deep Learning approach

Fig 3.2 demonstrates a general deep learning architecture in terms of an encoder-decoder framework, skip connections, and optional attention blocks.

C. Transfer learning Approach:

Transfer learning has proven to be a successful method of brain tumor classification using MRI, especially when conditions of limited annotated data are observed. VGG, ResNet and EfficientNet are pre-trained convolutional neural networks (CNNs) that are originally designed to run on large scale datasets (e.g. ImageNet), and are fine-tuned on medical imaging tasks. In a regular implementation, the convolutional layers are held at the lower levels in order to maintain the generic representations of features whereas higher levels are fine-tuned to acquire MRI-specific tumor properties. Reported classification accuracies [8], [12], [24] indicate a classification accuracy of between 98 and 99.66 indicating effectiveness of domain adaptation between natural images and medical imaging settings.

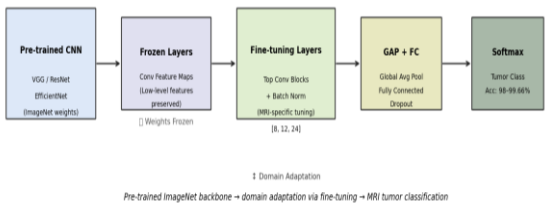

Fig 3.3 Transfer learning approach

Fig. 3.3 depicts a typical transfer learning system, comprising of a pre-trained backbone network, frozen feature extraction layers partially, fine-tuning blocks that are task specific and a final classification block.

D. Trend Synthesis and Survey Research:

Besides the model development research studies, there are also other studies that concentrate on synthesizing and analyzing the current research trends in the MRI-based brain tumor detection. The survey presented in [3] is a comprehensive overview of the deep learning methodologies, covering the advancements in the architecture and revealing the gaps in the research. A current systematic review in [18] reviews the literature in the timeframe 2020-24 years, which provides organized knowledge about the transfer learning strategies, transformer-based architecture, attention, and autoencoder-based approaches. A comparative benchmarking and cross-architecture performance analysis is introduced in [10], where an emphasis is placed on metrics of evaluation and on the generalization abilities of the models. In [11], hybrid CNN-Transformer segmentation frameworks are critically examined with a report of Dice similarity scores of greater than 0.90 over benchmark datasets. In addition, the context-aware and multimodal fusion approaches are slightly detailed in [13], which reiterates the significance of using complementary imaging modalities to increase diagnostic performance.

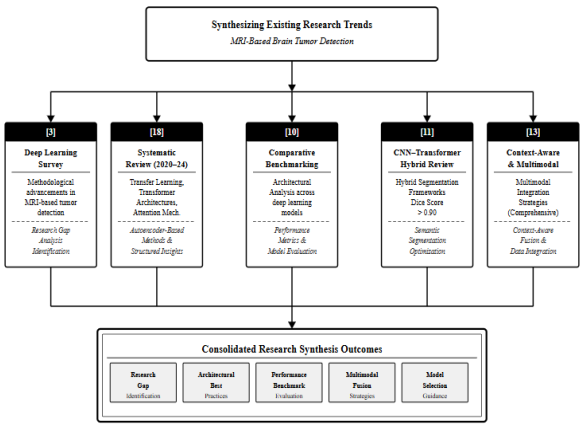

Fig 3.4 Synthesis Trend Survey Research

Fig. 3.4 displays the research synthesis framework, which summarizes groupings of thematic and consolidated research findings.

E. Hybrid and Attention Based Models:

Recently, hybrid networks, especially CNN-Transformer networks [11], [20], attention-based 3D U-Nets [19], [21], have been of considerable interest to the field of MRI-based brain tumor segmentation. The models combine the advantages of both Convolutional Neural Networks (CNNs) to grasp the local spatial fineness and the ability to model the global context of the target data by using Transformer encoders. The hybrid CNN-Transformer model in Figure 5 illustrates how the MRI input (2D or 3D slices) is first fed into a CNN backbone to extract hierarchical local features using the convolutional layers, pooling functions, and normalization operations. The extracted feature maps are tokenized through patch embedding and fed through a Transformer encoder. Transformer module uses multi-head self-attention and feed-forward networks to represent long-range relationships and global contextual relationships between tumor regions. The hybrid design provides a better precision in segmentation boundaries, detection of heterogeneous tumor sub-regions,

and uncertainty aware prediction mechanisms that facilitate better clinical deployment. CNN-Transformer architectures outperform CNN or Transformer models alone in difficult medical imaging applications by fusing their local spatial encoding and global semantic reasoning.

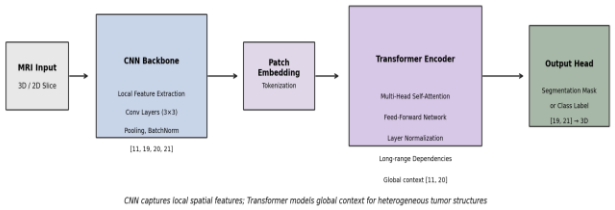

Fig 3.5 Hybrid attention based approach Model

F. Multi-task Prognosis Model:

Multi-task learning models learn to perform tumor segmentation, classification and predict prognosis jointly through shared encoder representations learnt by using the MRI inputs. Usually, a U-Net or CNN backbone can be used as a typical feature extractor, with segmentation, tumor subtype classification, and survival estimation typically having separate task-specific branches [17], [22]. The shared encoder as shown in Fig. 33.6 supplies parallel heads that are minimized by a joint loss function which increases the feature generalization and enables in-built clinical decision-making systems.

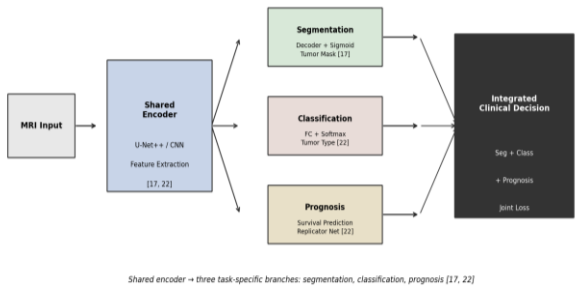

Fig 3.6 Multi-task Prognosis model

G. Real time detection Model:

Real time detecting models lay stress on quick localization of tumor to assist in time sensitive clinical practices. Architectures based on YOLOv7 [2] support end-to-end detection using a backbone network to process preprocessed slices of MRI scans and use non-maximum suppression to make predictions of bounding-boxes. This pipeline allows the efficient localization of tumors with competitive accuracy as shown in Fig. 3.7, so it can be included in the real diagnostic workflow.

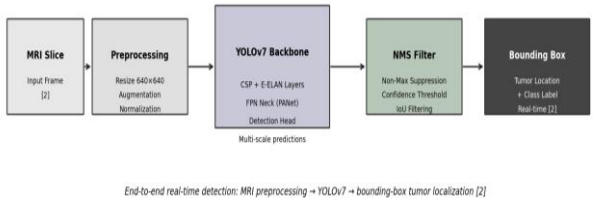

Fig 3.7 real time detection model

G. Context-aware multi model:

State-of-the-art context-aware models leverage multi-modal MRI, clinical, and genomic information to enable holistic characterization of tumors, moving beyond the boundaries of traditional segmentation evaluation criteria [13]. These models learn modality-wise features via modality-specific encoders and fuse them via techniques such as concatenation or attention-based alignment. As shown in the Fig. 3.8, the fused features are then processed by a hybrid predictive network to enable personalized diagnosis and treatment, thus improving clinical decision support systems.

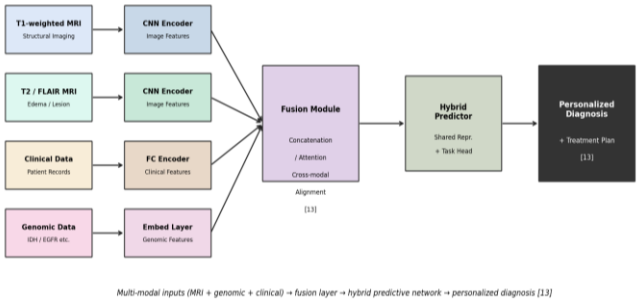

Fig 3.8 context-aware Multimodal

IV.COMPARATIVE TABLE

TABLE 1 COMPARATIVE ANALYSIS OF LEARNING-BASED MRI BRAIN TUMOR ANALYSIS METHODS

Ref	Task Focus	Methodology	Dataset	Metrics	Contribution
1	Classification	ML+ Feature Extraction	Public MRI	97%	Classical ML Feasibility
2	Detection + Classification	YOLO V7	Robo Flow MRI dataset	mAp=0.897	Real time tumor detect
3	Review	Deep Learning Survey	Multiple	Not applicable	Trend & gap analysis
4	Segmentation+ classification	DL Ensemble + Segmentation	MRI Dataset	Accuracy = 96%	Segmentation-driven classification
5	Classification	ResNet50+ Grad-CAM	MRI dataset	Accuracy = 98.52%	Explainable AI
6	Classification	ML Feature Optimization	Kaggle MRI dataset	Accuracy = 99%	Feature-Classifier
7	Classification	Segmentation + Ensemble CNN	MRI Dataset	Accuracy = 97%	Robust Ensemble based
8	Classification	Transfer Learning	MRI dataset	Accuracy = 98%	Small dataset adaptation
9	Binary Classification	ResNet 50 + Grad-CAM	Kaggle MRI (2024)	Accuracy = 98.52%	Binary Explainable diagnosis
10	Survey + Classification	Efficient + Dual CNN	Multiple	Accuracy = 98.5%	Benchmark feature

11	Review	CNN-Tranformer hybrid	Multiple	Dice > 0.90	Hybrid segmentation review
12	Classification	Transfer Learning	Br35H dataset	Accuracy = 99.66%	Small dataset optimization
13	Review	Context-aware ML	Multiple Dataset	Not applicable	Imaging-genetic integration
14	Classification	Optimized CNN	MRI	Accuracy = 97%	Hyper parameter tuning
15	Classification	ML-texture Feature extraction+ML Classifier	Figshare MRI	Accuracy = 99.4%	Texture-based ML approach
16	Classification	Shallow Attention CNN	Cheng MRI dataset	Accuracy = 98.4%	Edge deployable Model
17	Segmentation + Prognosis	U-Net+++ ML+PSO	BraTS dataset	AUC = 0.70	Survival Prediction
18	Systematic Review	Transfer, Autoencoder, Transformer	60 studies	Not applicable	DL Trend analysis
19	Segmentation	Calibrated 3D attention U-Net	BraTs 2020	Dice = 0.912	Uncertainity Calibration
20	Segmentation	3D CNN + Transformer	Multi modal MRI	Dice = 0.916	Global-Local feature fusion
21	Joint Learning	Deformable Attention + XAI	MRI Dataset	>96.5% accuracy	Multi-task framework
22	Segmentation + prognosis	CNN+ Replicator NN	MRI Dataset	High reliability	Outcome prediction
23	Classification	Ensemble DL + XAI	MRI Dataset	98% accuracy	Stability & explainability
24	Classification	Fine-tuned Transfer learning	MRI Dataset	99% accuracy	Feature-level refinement
25	classification	Hybrid Feature	MRI Dataset	98.8% accuracy	Feature Interpretability

Table 1 supports the fact that the development of classical machine learning with handcrafted features has moved to deep learning and hybrid CNN-Transformer models to analyze brain tumors based on MRI. Classical ML has a high accuracy of 97-99 percent, but does not have the robustness, whereas deep learning and transfer learning are always more than 98 percent and Dice >0.90 in segmentation. Hybrid and attention models enhance the segmentation, classification and prediction of prognosis, as well as better delineation of tumor boundaries. Lightweight and real-time frameworks make clinical deployment, but less integrated explainable multi-task frameworks are an open research direction.

V. COMPARITIVE INSIGHTS AND RESEARCH GAPS

The comparison allows identifying a distinct methodological development between the handcrafted feature-based machine learning models to the sophisticated deep learning and hybrid architectures. Classical ML methods show competitiveness in controlled environments; nevertheless, they are dependent on manual features engineering, scaling and generalization are restricted. By comparison, the deep learning models, specifically the U-Net variant, attention-based networks and CNN-Transformer hybrids, are better at segmentation tasks, often scoring above 0.90 on any benchmark dataset, e.g. BraTS. Transfer learning procedures have proven to be highly accurate in terms of classification (up to 98 percent) in low-data conditions, although numerous articles are based on single-source databanks, which casts doubt on cross-dataset validity and clinical applicability. Even though segmentation and classification tasks are well studied, any multi-task framework that incorporates segmentation, diagnosis, prognosis prediction, and explainability has not been studied in detail. Also, in spite of the growing use of explainable AI methods and uncertainty-sensitive models, there is a lack of real-time clinical implementation and validation in various patient populations. The above observations demonstrate that there is a need to have integrated, computationally efficient and clinically reliable, frameworks, which combine segmentation accuracy, interpretability, and outcome prediction in a single system.

VI. FUTURE RESEARCH DIRECTION

Although great progress has been made in multimodal MRI-based brain tumor segmentation, there are a number of issues to overcome before it can be successfully used in clinical practice. Despite the fact that the frameworks presented in [19] and [20] exhibit a high accuracy at the voxel level and allow enhanced delineation of the tumor using the latest deep learning based architectures, their implementation in the real time clinical setting remains a limitation due to their computational complexity and the hardware cost. Future studies should thus focus on lightweight and memory efficient implementations to make low-latency inference possible on a variety of clinical settings. Moreover, although [20] demonstrates that hybrid CNN-Transformer models are efficient in terms of enabling the enhancement of global-local feature fusion, further research on scalable multimodal attention mechanisms is required to better represent tumor heterogeneity. It is also necessary to expand the existing segmentation-oriented models to multi-task based learning systems that include classification, prognosis prediction, and uncertainty estimation. Lastly, enhancing model interpretability and performance improvement on smaller or irregular tumor subregions are also critical lines of attack towards clinical reliability of systems.

VI. DISCUSSION AND ANALYSIS

The literature reviewed illustrates a distinct transformation of the conventional machine learning methods to the sophisticated deep learning methods of brain tumor analysis by MRI. The original methods [1], [6], [15] were mainly based on manual feature extraction with traditional classifiers, with satisfactory accuracy of classification, but poor generalization with non-homogenous data. Convolutional neural networks (CNNs) were introduced, and

this created a big breakthrough in terms of robustness because these models can learn hierarchical features automatically using multimodal MRI data. CNN-based models optimized using segmentation to a greater level [4], [7], [14] improved structural definition and diagnostic accuracy. In order to mitigate the problems of data scarcity, transfer learning plans [8], [12], [24] were highly popular to significantly enhance the performance of the limited-data problems with a few studies reporting a classification accuracy over 99%. More recently, hybrid CNN-Transformer designs [19], [20] have become effective with respect to modeling long-range correlations and bettering tumor boundary delineation, often recording Dice similarity scores greater than 0.90. In addition to segmentation, multi-task learning systems [17], [22] can be also applied to survival estimation and clinical outcomes analysis and explainable AI systems [5], [9], [23], [25] are more transparent and can be interpreted by clinicians. Moreover, lightweight and real-time models of detection [2], [16] help to move towards the practical implementation in the working process of hospitals.Irrespective of these advancements, a research gap is still evident in the literature: a lack of a harmonized, computationally efficient framework, which will smoothly combine segmentation, classification, prognosis prediction, and interpretability into a unified clinically deployable system.

VII. CONCLUSION

The undertaking of this survey offers an in-depth review of learning-based methods in the analysis of MRI brain tumors, the classical machine learning, deep learning, hybrid architectures, and explainable AI-based approaches. The review demonstrates a vivid step forward into automated hierarchical representation learning and greater accuracy of segmentation with the development of sophisticated deep learning frameworks over the course of handcrafted features. Transfer learning and ensemble-based methods have been found to improve the strength of limited-data context, and attention processes and CNN Transformer models have been developed to improve the delineation of tumor boundaries further. Additionally, explicable AI practices help to provide increased transparency and confidence in clinical practices.

Though these developments have been done, there are still significant challenges, such as the real-time clinical implementation, the strong multi-task prediction, and the correct delineation of small or heterogeneous tumor subregions. The literature shows that there is a necessity of computationally efficient, united frameworks, combining segmentation, classification and the prediction of prognosis by integrating interpretability and uncertainty-conscious modelling. The challenges will have to be overcome to translate research accomplishments to reliable, clinical deployable diagnostics systems.

V. REFERENCES

[1] Saeedi et al., “MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning technique,” BMC Medical Informatics and Decision Making, vol. 23, 2023, doi: 10.1186/s12911-023-02114-6.

[2] R. Nimmagadda and P. K. Devi, “A deep learning approach for brain tumour classification and detection in MRI images using YOLOv7,” Frontiers in Oncology, vol. 15, Art. no. 1508326, 2025, doi: 10.3389/fonc.2025.1508326.

[3] T. A. Fahim, F. B. Alam, and M. A. Hossain, “Brain tumor detection, classification and segmentation by deep learning models from MRI images: Recent approaches, challenges and future directions,” 2025.

[4] S. Mohsen, S. Oraby, and M. Abdel-Aziz, “Deep learning and machine learning for brain tumor detection: A review, challenges, and future directions,” Archives of Computational Methods in Engineering, 2025, doi: 10.1007/s11831-025-10416-3.

[5] M. M. Mohamed, M. T. R. Mahesh, V. K. Vinoth, and S. Guluwadi, “Enhancing brain tumor detection in MRI images through explainable AI using Grad-CAM with ResNet-50,” BMC Medical Imaging, 2024, doi: 10.1186/s12880-024-01292-7

[6] T. Berghout, “The neural frontier of future medical imaging: A review of deep learning for brain tumor detection,” Journal of Imaging, vol. 11, no. 1, Art. no. 2, 2025, doi: 10.3390/jimaging11010002.

[7] M. Ashimgaliyev, B. Matkarimov, A. Barlybayev, R. Y. M. Li, and A. Zhumadillayeva, “Accurate MRI-based brain tumor diagnosis: Integrating segmentation and deep learning approaches,” Applied Sciences, vol. 14, no. 16, Art. no. 7281, 2024, doi: 10.3390/app14167281.

[8] K. Kumar, K. Jyoti, and K. Kumar, “Machine learning for brain tumor classification: Evaluating feature extraction and algorithm efficiency,” Discover Artificial Intelligence, vol. 4, Art. no. 112, 2024, doi: 10.1007/s44163-024-00214-4.

[9] V. R. Srinivas and R. Parvathi, “Explainable AI-driven MRI-based brain tumor classification: A novel deep learning approach,” Frontiers in Artificial Intelligence, vol. 8, Art. no. 1700214, 2026, doi: 10.3389/frai.2025.1700214.

[10] G. M. Salama, M. K. Abd-Ellah, A. I. Hussein, E. S. Bayoumi, S. Ashraf, and M. Safy, “Brain tumor diagnosis techniques: Key achievements, lessons learned, and a new CNN architecture,” Egyptian Journal of Radiology and Nuclear Medicine, 2025, doi: 10.1186/s43055-025-01567-1.

[11] A. Hamza and R. Damaševičius, “Deep learning for brain tumor segmentation and classification: A systematic review of methods and trends,” Computers, Materials & Continua, 2025, doi: 10.32604/cmc.2025.069721

[12] A. B. Naeem, B. Senapati, and A. Zaidi, “Enhancing brain tumor detection from MRI-based images through deep transfer learning models,” AI, vol. 6, Art. no. 305, 2025, doi: 10.3390/ai6120305.

[13] U. Amjad, A. Raza, A. Akhunzada, M. Fahad, M. Abubakar, D. Farid, and H. Beenish, “Context-aware machine learning techniques for brain tumor classification and detection—A review,” Heliyon, 2025, doi: 10.1016/j.heliyon.2025.e41835.

[14] M. Aamir, A. Namoun, S. Munir, N. Aljohani, M. H. Alanazi, Y. Alsahafi, and F. Alotibi, “Brain tumor detection and classification using an optimized convolutional neural network,” Diagnostics, vol. 14, no. 16, Art. no. 1714, 2024, doi: 10.3390/diagnostics14161714.

[15] H. K. Mishra, G. K. Jha, and K. Mishra, “Automated primary brain tumors classification of MR images using texture features extraction and machine learning,” Int. J. Intell. Syst. Appl. Eng., vol. 12, no. 22s, pp. 2244–2248, Aug. 2024.

[16] N. A. Babar, J. Lateef, S. Syed, J. Dietlmeier, N. E. O’Connor, G. B. Raupp, and A. Spanias, “Brain tumor classification in MRI scans using edge computing and a

shallow attention-guided CNN," Biomedicines, vol. 13, Art. no. 2571, 2025, doi: 10.3390/biomedicines13102571.
[17] R. Srikanth, N. Kanya, and P. S. Rajakumar, "Detection and classification of brain tumours from MRI images with prediction of the overall survival rate in glioblastoma using machine learning techniques," International Journal of Intelligent Systems and Applications in Engineering, 2024.
[18] S. Bouhafra and H. El Bahi, "Deep Learning Approaches for Brain Tumor Detection and Classification Using MRI Images (2020 to 2024): A Systematic Review," Journal of Imaging Informatics in Medicine, vol. 38, no. 3, pp. 1403–1433, Jun. 2025, doi: 10.1007/s10278-024-01283-8.
[19] V. R. Yanamadni and H. Bommala, "Reliable Brain Tumor Segmentation Using CalibSeg: A Temperature-Calibrated 3D Attention-UNet with Uncertainty Quantification," 2025.
[20] V. R. Yanamadni and H. Bommala, "Lightweight 3D Hybrid U-Net–Transformer Architecture for Accurate Multimodal MRI Brain Tumor Segmentation," 2025.
[21] E. Zarenia, A. A. Far, and K. Rezaee, "Automated multi-class MRI brain tumor classification and segmentation using deformable attention and saliency mapping," Scientific Reports, vol. 15, Art. no. 8114, 2025, doi: 10.1038/s41598-025-92776-1.
[22] D. Rastogi, P. Johri, M. Donelli, S. Kadry, A. A. Khan, G. Espa, P. Feraco, and J. Kim, "Deep learning-integrated MRI brain tumor analysis: Feature extraction, segmentation, and survival prediction using replicator and volumetric networks," Scientific Reports, vol. 15, Art. no. 1437, 2025, doi: 10.1038/s41598-024-84386-0.
[23] K. M. Hosny, M. A. Mohammed, R. A. Salama, and A. M. Elshewey, "Explainable ensemble deep learning-based model for brain tumor detection and classification," Neural Computing and Applications, vol. 37, pp. 1289–1306, 2025, doi: 10.1007/s00521-024-10401-0.
[24] D. Rastogi, P. Johri, M. Donelli, L. Kumar, S. Bindewari, A. Raghav, and S. K. Khatri, "Brain Tumor Detection and Prediction in MRI Images Utilizing a Fine-Tuned Transfer Learning Model Integrated Within Deep Learning Frameworks," Life, vol. 15, p. 327, 2025, doi: 10.3390/life15030327.
[25] A. Rahman, M. Hayat, N. Iqbal, F. K. Alarfaj, S. Alkhalaf, and F. Alturise, "Enhanced MRI brain tumor detection using deep learning in conjunction with explainable AI SHAP based diverse and multi feature analysis," Scientific Reports, vol. 15, p. 29411, 2025, doi: 10.1038/s41598-025-14901-4.